Olivier Berg

A12748H1

COLLECTION PAPER

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MACROECONOMICS

TRINITY TERM 2026

Please start the answer to each question on a separate page.

There are FIVE questions in this paper.

Answer **ALL** short questions from Part A and **ALL** long questions from Part B.

Part A attracts 40% of the marks in total and within part A all questions carry equal weight; each question in part B attracts 30% of the total marks.

Candidates may use a calculator as specified by the Department of Economics.

Do not turn over until told to do so.

PART A

Answer **ALL** three questions in this section.

1. Consider a closed economy *IS-PC-MR* model. There is a one period lag in the effect of real interest rates on output in the *IS* curve. The Phillips Curve is based on adaptive expectations for inflation, and equilibrium output depends on equilibrium employment determined in the *WS-PS* model. The economy starts in equilibrium. Suppose that in period t (and in advance of real interest rates being set for period t) there is a positive and unexpected cost-push shock that lasts for one period only.

   (a) Explain the adjustment of real interest rates, output and inflation from period t until the economy reaches equilibrium. [30%]

   (b) Explain how the inflation deviation from target in period $t+1$ is related to the slope of the Phillips Curve. Provide economic intuition for your answer. [35%]

   (c) Define involuntary unemployment and explain the path followed by involuntary unemployment from period t until the economy reaches equilibrium. [35%]

2. Borduria (B) and Syldavia (S) are two countries described by independent Solow growth models. They are characterised as follows:

	Borduria (B)	Syldavia (S)
Production function	$Y_B = 20K_B^{1/2}L_B^{1/2}$	$Y_S = 10K_S^{1/2}L_S^{1/2}$
Savings rate	$s_B = 0.1$	$s_S = 0.2$
Depreciation rate of capital	$\delta_B = 0.1$	$\delta_S = 0.1$

   Population and technology are constant in each country.

   (a) Calculate the steady-state values of capital per capita and output per capita in each country. [30%]

   (b) Is it possible that the people of Borduria sometimes have a lower per capita income than those of Syldavia? Why or why not? [20%]

   The people of Syldavia want to have the same steady state per capita income as those in Borduria.

   (c) By how much must the people of Syldavia change their savings rate to achieve the same steady state per capita income as Borduria? Discuss the magnitude of the change needed and the implications for Syldavia's capital to labour ratio. [30%]

   (d) Assuming that Syldavia was initially in steady state, graphically represent the adjustment to the new steady state after the people of Syldavia raise their savings rate to the level calculated in part (c). [20%]

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3. Consider the Hall Random Walk model of consumption.

(a) State the assumptions of the model, and give a mathematical derivation and an intuitive explanation of its main implications. [40%]

(b) Suppose the assumptions of the model are satisfied. In each period income is given by the constant, y, minus a loss, L, that occurs with probability $p \in (0,1)$. The loss is incurred before a household decides their consumption for the period. That is, consumption is determined knowing whether the loss has been incurred in the current period, but only knowing the probability of incurring it in future periods. Assume a household starts time 0 with assets A_0; show that if the household does not incur the loss then

$$c_0 = rA_0 + y - \frac{pL}{1+r},$$

whereas if it incurs the loss

$$c_0 = rA_0 + y - \frac{(r+p)L}{1+r}.$$

Interpret these results. [30%]

(c) Suppose that a household's utility function is logarithmic, so that they maximise

$$\sum_{t=0}^{+\infty} \beta^t \ln c_t,$$

where β is the subjective discount factor. Explain how you would expect c_0 in part (b) to change. You do not need to find a mathematical expression for the new level of consumption, but can instead illustrate your answer with a diagram showing the marginal utility of consumption. [30%]

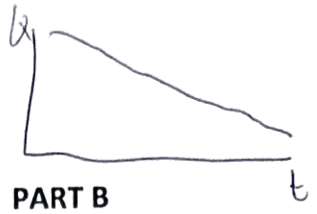
$\sum_{i=0}^{\infty} a^i = \frac{1}{1-a}$

$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} \ldots = \frac{1}{2/3} = 1\frac{1}{2}$

$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots = 2$

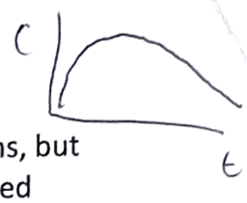
$\frac{1}{1-\left(\frac{1}{1+r}\right)} = \frac{1}{\left(\frac{1+r-1}{1+r}\right)} = \frac{1+r}{r}$

$\left(\frac{1}{1+r}\right) c_0$



PART B

Answer **ALL** questions in this section.



4. "Sticky prices in goods markets can be explained from optimising behaviour by firms, but the macroeconomic predictions of sticky price models are inconsistent with observed macroeconomic behaviour." Discuss.

5. In order to match the observed data on business cycle fluctuations, early calibrations of the RBC model relied on measures of the Solow residual implying high volatility and high persistence of total factor productivity, as well as a high probability of technological regress. Discuss why this approach leads to challenges to the validity of RBC theory.

$$\frac{\partial U(C_t, L_t)}{\partial C_t} = \delta \frac{\partial U(C_{t+1}, L_{t+1})}{\partial C_{t+1}}$$

$$\frac{\partial U(C_t, L_t)}{\partial C_t} = \delta \frac{\partial U(C_t, L_t)}{\partial L_t} \frac{1}{w_t}$$

$$\frac{\partial U(C_t, L_t)}{\partial L_t} = \delta \frac{\partial U(C_{t+1}, L_{t+1})}{\partial L_{t+1}} \frac{w_t}{w_{t+1}}$$